TECHNICAL SPECIFICATION

DIGITAL TELE PROTECTION SIGNALLING EQUIPMENT

Design, manufacture, factory testing before dispatch, packing, supply, installation, testing and commissioning of Digital Tele Protection (Couplers) Signalling Equipment (DTPC) on the EHT line sections.

Integration:

It is the total responsibility of the bidder to integrate the entire supplied equipment with the existing communication system (OLTE). However the bidder should supply any necessary cables, connectors etc. required for integration without any additional cost.

Cabling:

All signalling cables within the telecommunication system, and the power supply cables are part of contractor's delivery and installation. Signal cables and power supply cables shall be placed apart from each other (minimum distance 150 mm). Redundant cables and cables to redundant equipment shall be placed in separate cable routes. All Contractors supplied inside plant cables and cable assemblies shall be constructed from non-PVC, fire/smoke resistant materials.

Whether called for specifically or not, all accessories (attenuators, patch cards, test probes etc.)required for normal and satisfactory operation (as deemed fit by the purchaser) shall be considered to be a part of the tenderers basic scope of supply and no claim for extra payment will be accepted on these grounds.

Supply, Laying and Termination with proper lug system of AC and DC Cables and earthing of Digital tele- protection equipment with 7/20 wire is in Bidders scope. 25Mtrs of 2 (two) Core 2.5 Sq.mm multi strand unarmoured Copper Cable for AC for each cabinet and DC Supply cable for source-1 (25mtrs) DC supply cable for source-2 (25mtrs) required with each DTPC and 10 Mtrs of 7/20 wire is required per cabinet for earthing system and earthing shall be done with proper lug system from nearest earth point. 230V AC power supply is required at least one point in the cabinet/panel.

Power Supply Redundancy:

The Digital Tele Protection equipment shall have two power supplies to facilitate power supply redundancy. The output of both the power supplies are shorted together on the backplane and then in turn given to all the cards.

There shall be two power supply slots located on the chassis for equipping the power supply modules.

Power supply Operating voltages**:**-48VDC± 10%
Protection signalling equipment (compatible to fibre optic equipment)

The protection signaling equipment is intended for providing (i) inter-tripping (ii) direct tripping(iii) Permissive Tripping (iv) Blocking commands of HV lines.

Communication Interface / Channel redundancy:

The DTPC should have the redundant communication interface, i.e.,: E1 plus Ethernet ports for providing diversified E1 plus Ethernet communication channel over alternate optical link in addition to direct point to point main / default E1 with automatic failover protection channel. The redundant E1 plus Ethernet ports may be provided on different modules if possible.

DTPC management:
The interconnected DTPCs should be accessible & manageable from one end to other end and vice-versa as long as the E1 plus Ethernet channels are healthy. Further, the DTPC management port shall be accessed from any remote location on the Ethernet over SDH/MPLS-TP network.

Clock synchronization:
The DTPC clock should be able to get synchronized with local SDH /MPLS-TP clock through the E1 port plus Ethernet port with which the DTPC is interfaced. The other option of having clock from the local / remote GPS also may be provided and manual clock settings have to be provided as default.

Digital protection signaling equipment shall be suitable for communicating through fibre optic equipment.

Digital Tele-protection coupler equipment shall be integrated to the existing OLTE on E1 plus Ethernet interface. Digital Tele-protection equipment shall be provided with visual trip counters, event registers to register events which can be down loaded in to PC or Laptop.

The protection signaling equipment shall have digital electrical output on E1 plus Ethernet Interface. Bidder shall quote for protection signaling equipment suitable for 8 commands with trip counters, event registers in each equipment and suitable for transmission on E1 interface/Ethernet interface.

High security and dependability shall be ensured by the contractor. Probability of false trippingandfailuretotripshallbeminimum.Curves/figuresindicatingabove-mentionedmeasures shall be submitted with the tender proposal.

The routine and type tests of the protection signaling equipment shall be as per the IEC recommendations and suggestions of the latest revision**.**

Principle of operation:
During normal operation protection signaling equipment shall transmit a secured guard code. In case protection signaling equipment receives one or more command it should interrupts the guard code and shall transmit the command code in sequence. The receiver recognizes the command code and absence of the guard code and generates the command.

All signals processing i.e., the generation of tripping signal and the evaluation of the signals being received shall be performed completely digitally by using DSP. The digital techniques thus employed obviate any need for calibration.

Loop testing:
An automatic loop testing routine shall cyclically check the Tele-protection channel. The test signal shall be transmitted in the same way as the genuine tripping signal shall be recognized by the receiver and shall be reflected back to the transmitter. In case test signal does not reach the transmitter, the test should be repeated automatically for at least 5 trials and alarm should be given in case the test signal fails.

It shall also be possible to initiate a loop test manually at any station by pressing a button on the front of the equipment.

Internal test routine shall continuously monitor the availability of the protection signaling equipment.

Proper tripping signal shall always take the priority over the test procedure.

The high speed digital protection signaling equipment shall be designed and provided with following features.

- Shall employ latest digital signal processing technology.
- Shall be provided with redundant 48V DC power supply unit
- It should communicate on 2Mbps (E1) plus 10/100Mbps (Ethernet) channel.
- All I/Os electrically isolated
- Full duplex operation
  Permissive tripping
  Direct tripping
- Blocking, unblocking
- Auto loop test facility shall be provided
- Should meet **IEC834-1standard for E1**
- Should meet **IEEE 802.3standard for Ethernet**.
- Shall provide an interface facility to connect handheld terminal or PC for indicating status, alarm messages and transmission time etc.
- Shall have the addressing facility
- Shall able to transmit up to 8commands in oneE1 / Ethernet channel
- Transmission time shall be<7ms.
- The Tele protection equipment shall be programmed through handheld terminal/ PC.
- The Tele-protection equipment shall have a visual counter.
- Visible counters for each command (Input/ Output) should be provided.
- Proper backup facility (Non-volatile) should be available to register at least 7500 events/alarms up to 65K counts for each command/test to be stored in the Digital Tele-protection coupler and should be accessed through Remote login facility.
- All the Digital Protection Coupler should work as standalone equipment.

Protection signaling equipment design

The Tele-protection equipment shall be of modular construction and shall be an integrated to the existing OLTE, and optical terminal multiplexer equipment and shall be completely solid-state using semiconductors, micro-processors and self-supervision, providing alarms in the case of equipment failure (TX / RX), absence of the guard frequency signal, power supply failure, etc.

The input/output interface to the protection equipment shall be by means of relays and the input/output rack wiring shall be carefully segregated from other shelf/cubicle wiring.

The isolation requirements of the protection interface shall be for 2kv rms. Each tele-protection equipment shall be supplied with redundant 48V DC power supply unit.

All command channels are operationally independent from each other and they meet the following performance characteristics:

. equipment operating time<6 ms(8channels)
. probability of missing command p_{mc}(at ber<10-3) <10-9
. mean time between unwanted commands
(at t_{ses}<0.005% and t_{es}<0.1%) 100000a

The error control procedure in the 2.048 Mbps (E1/Ethernet) 2Mbps channel is based on the use of a block code specifically developed to be used for tele-protection signaling via tele-communication networks based on the ITU-T recommendations. The error control procedure includes a means to identify the origin of the incoming 2Mbps.

Command-type Tele-protection applications are normally placed in three categories according to the nature of the information conveyed by the Tele-protection commands: permissive, blocking and inter-tripping schemes.

The Tele-protection signaling equipment shall work as standalone equipment and shall be accessed through Remote login through via OLTE Network.

The Tele-protection equipment shall be provided with trip counters for both TX and RX commands in the form of event register.

Main characteristics:

2Mbps interface

The 2Mbps interface meets the requirements of the ITU-T recommendation G.703.6.

a. Electrical Interface : G.703(E1)
b. Bit rate: 2048Kbps +/- 50 ppm
c. Nominal Impedance 120 Ohms balanced& interface of krone type.

10/100 Mbps interface

The 10/100Mps interface meets the requirement of the ITU-T recommendation.

a. Electrical Interface : RJ45
b. Bit rate :10/100Mbps
c. Nominal Impedance 100 ohms.

Command interfaces:

Command inputs: 8

The command inputs should be galvanic ally isolated inputs, which generate internally the operating voltage for the input circuit.

a) Potential free contacts
b) Operating current 5 ma
c) Command Voltages : operating voltage 220V DC-250V DC and voltages of 45V-60V DC end to end coupler testing shall be provided.
d) Minimum operating command voltage: 170-175V DC for 220V DC & 40-45V for 48V DC (for testing)

Command outputs: 8

The command outputs should be galvanic ally isolated outputs. Each output has one MOS-FET change over contact**.**

a) Nominal voltage : operating voltage 220V DC-250V DC and voltages of 45V-60V DC end to end coupler testing shall be provided.
b) Maximum switching power 100 VA
c) Maximum switching current 5A
d) Other parameters: As per IEC-255-0-20

Command interface-related requirements:

All interfaces with the protection relay logic and command system shall comply with the standards of IEC255/IEC834-1/IEE 802.3 and also other standards applicable to DTPC.

GUARANTEED TECHNICAL PARTICULARS FOR DIGITAL TELEPROTECTION COUPLER (DTPC) EQUIPMENT:

1. Make & Model :
2. Commands capacity :
3. Compatible to Optical Equipment :Yes /No :
4. Type of Communication Interface :
5. Type of Command Input Interfacing :
6. Type of Command output Interfacing :
7. No. of commands capability :
8. Type of Trip counters (event registers) :
9. Power Supply redundancy : Yes/No
10. Operating time for Protection couplers :
11. Interfacing through Potential free contacts : Yes/No
12. Monitoring through LAN Network /wideband Network via OLTE equipment for the DTPCs offered :
13. Whether the offered DTPCs are an integral part of OLTE eqpt.:
14. Visible counters for each command(Input/ Output) provided: Yes/No.

Signature of the tenderer

List of mandatory spares for Digital Tele-protection equipment

SL. No.	DESCRIPTION	Quantity
1.	CONTROLUNITCARDS	
2.	POWERSUPPLYCARDS	
3.	Channel Interface card.	
4	Any other cards needed to meet spec. requirement as a mandatory spare	

Date: **Signature:**

Place: **Name:**

Seal: